

## CY 545: Reagents in Organic Synthesis

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| Course Code      | CY 545                                                                         |
| Course Name      | Reagents in Organic Synthesis                                                  |
| L-T-P-C          | 3-0-0-3                                                                        |
| Prerequisites    | Basic Knowledge in organic reactions and mechanism and Instructor's permission |
| Intended for     | BS/M.Sc. Chemistry and Ph.D. (any discipline)                                  |
| Mutual Exclusion | None                                                                           |
| Approval         | 60th BoA (Jun 2025)                                                            |

### Preamble

This course, “Reagents in Organic Synthesis,” is designed to provide BS, M.Sc. and Ph.D. students with a deep and systematic understanding of the reagents that drive organic transformations. Going beyond memorization, the course emphasizes mechanistic insight, selectivity principles, and the strategic deployment of reagents in synthesis planning. From classical redox reagents to modern transition-metal complexes, from functional group interconversions to asymmetric catalysis, this course integrates both foundational knowledge and cutting-edge developments. Ultimately, this course aspires to empower students not just to understand how reagents function, but to master how to select and apply them creatively in complex synthetic contexts.

### Course Modules

- **Module 1. Oxidation Reactions in Organic Synthesis:** (8 Hours)
  - *Chromium-Based Oxidizing Reagents:* Chromic acid (Jones reagent), PCC (Pyridinium chlorochromate), PDC (Pyridinium dichromate), Collins reagent: Preparation, oxidation of primary/secondary alcohols, Selectivity issues: overoxidation, functional group compatibility, Etard reaction
  - *DMSO-Based Oxidizing Reagents:* Moffatt Oxidation, Pfitzner–Moffatt oxidation, Parikh–Doering Oxidation, Albright–Goldman Oxidation, Swern Oxidation, Corey–Kim Oxidation, Kornblum oxidation
  - *Oxidation using Organic Peroxide/Peracid:* Baeyer–Villiger oxidation, Dakin reaction, Prilezhaev epoxidation, Dioxirane
  - *Other Oxidizing agents:* CAN,  $\text{SeO}_2$ ,  $\text{MnO}_2$ , Fetizon's reagent, TPAP-NMO, IBX, and Dess–Martin periodinane,
  - *Asymmetric Epoxidation and Dihydroxylation:* Sharpless Asymmetric Epoxidation, Sharpless Asymmetric Dihydroxylation, Jacobsen–Katsuki Epoxidation, Shi epoxidation
- **Module 2. Reduction Reactions in Organic Synthesis:** (8 Hours)
  - *Metal Hydrides:*  $\text{LiAlH}_4$ ,  $\text{NaBH}_4$ ,  $\text{NaCNBH}_3$ , Red-Al, DIBAL-H,  $\text{Me}_4\text{NBH}(\text{OAc})_3$ ,  $\text{Et}_2\text{BOMe-NaBH}_4$ ,  $\text{LiBH}_4$ , Luche's reagent, L and K-Selectrides, Superhydrides,  $\text{BH}_3$
  - *Single electron reduction:* Reduction of aromatics: Birch reduction, mechanism and regioselectivity, Reduction of alkynes and enones
  - Organic reducing agents like Hantzsch esters and relate their mode of action to biomimetic oxidation and reduction by NADP/NADPH.
  - *Asymmetric Reduction:* Alpine borane, CBS

- **Module 3. Boron-Based Reagents in Organic Synthesis:** Introduction to boron chemistry, Hydroboration: mechanism and stereoselectivity, Hydroboration and oxidation, Carbonylation and Cyanidation, Asymmetric Allylation and Crotylation: Brown approach, Roush approach (6 Hours)
- **Module 4. Palladium-Based Reagents in Organic Synthesis:** (6 Hours)
  - *Palladium-Catalyzed C–C Bond Forming Reactions:* Negishi coupling, Kumada coupling, Suzuki–Miyaura coupling, Stille Coupling, Hiyama coupling, Inter and Intramolecular Heck reactions, Sonogashira coupling, Buchwald–Hartwig amination, Pd-catalyzed carbonylation and cyanation, Tsuji-Trost reaction
  - *Palladium-Catalyzed Oxidation Reactions:* Wacker oxidation, Saegusa-Ito oxidation
- **Module 5. Copper-Based Reagents in Organic Synthesis:** (4 Hours)
  - Introduction to copper chemistry, Reactions of organocopper reagents with unsaturated carbonyl compounds, acid chloride, epoxides, and vinyl halides
  - Copper-catalyzed cross-coupling reactions: Cadiot-Chodkiewicz coupling, Glaser coupling, Castro-Stephen's reaction, Ullman coupling, Chan-Lam coupling
- **Module 6. Phosphorus/Sulfur-Based Reagents in Organic Synthesis:** (5 Hours)
  - *Name reactions involving Phosphorus Reagents:* Staudinger reaction, Corey-Fuchs reaction, Appel reaction, Mitsunobu reaction
  - *Phosphorus Ylides and the Wittig Reaction:* Formation and stability of phosphorus ylides, Stabilized vs unstabilized ylides: product control (Z/E selectivity), Schlosser modification of the Wittig reaction
  - *Sulfur Ylides or Sulfur based reagents:* Julia olefination and modification, Corey-Chaykovsky Reaction
- **Module 7. Chiral Auxiliaries in Asymmetric Synthesis:** Introduction to aldol chemistry, Formation of E and Z-enolates, Diastereoselective aldol reaction: Zimmerman–Traxler transition state, Asymmetric aldol reactions using Evans auxiliary, Crimmins auxiliary. Asymmetric alkylation of carbonyl compounds: Evans auxiliary, SAMP/RAMP. Application of asymmetric-aldol reactions in the synthesis of complex natural products. (5 Hours)

## Text Books

1. *Reagents for Organic Synthesis* by Fieser & Fieser (Vols. I–XIV).
2. *Strategic Applications of Named Reactions in Organic Synthesis* by Kurti & Czako, Elsevier Academic Press, 1st Edition, 2005.
3. *Modern Organic Synthesis: An Introduction* by Zweifel, Nantz, and Somfai, Wiley, 2nd Edition, 2017.

## References

1. *Advanced Organic Chemistry: Reactions and Synthesis* by Carey & Sundberg (Part A and B), Springer, 5th Edition, 2008
2. *Modern Methods of Organic Synthesis* by W. Carruthers & I. Coldham, Cambridge University Press, 4th Edition, 2004
3. *Organic Chemistry* by Clayden, Greeves, and Warren, Oxford University Press, 2nd Edition, 2012.